

Technical Report Direct Marketing Optimization

1. Task and data

The bank can contact 15% of its client base with one of three offers, consumer loan (CL), credit card (CC), or mutual fund (MF), and wants to maximize expected revenue. Labels (**Sale_***, **Revenue_***) are known for 969 of 1,615 clients (60%, **labeled set**); the remaining 646 (40%, **targeting set**) are unlabeled and are the clients to be selected as contacts.

Four source tables were joined using **Client** as key, socio-demographics, product holdings and balances, inflow/outflow (flows), and sales/revenues (labeled clients only, n/a for the unlabeled).

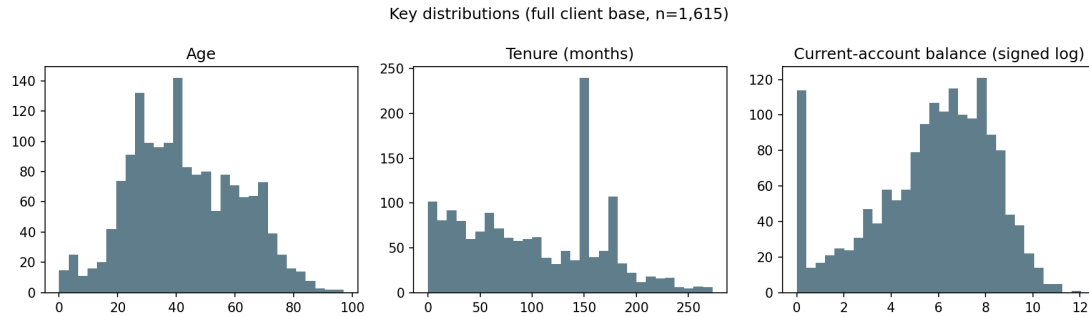


Figure 1: Distributions for Age, Tenure and Current Account

Figure 1. Its possible to see that Age is centered around the 20 to 40 year threshold, the Tenure shows a heavy kurtosis around the 150-152 months, and that the current account balance is skewed to the left, thus showing a left tail

Regarding the sell rate distribution while deep-diving into the labelled set, it was possible to find that the three products behave differently, as the Customer Loan is clearly dominant:

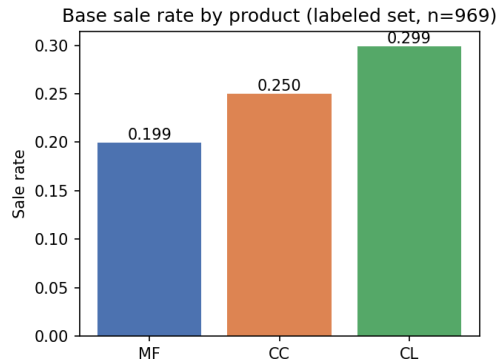


Figure 2: Base Sell Rate per Product

Figure 2. Customer Loan is the product with the highest base sell rate in the labelled set @ 0.299, whilst Mutual Fund is the lowest @ 0.199

2. Data quality and missing values

Table	Missingness	Diagnosis	Treatment
Products (<code>Count_*</code> , <code>ActBal_*</code>)	NaN where a product isn't held	Structural, not missing data	Filled with 0; added <code>has_SA/MF/OVD/CC/CL</code> ownership flags
Soc_Dem Sex	3 of 1,615	Negligible	Kept as a distinct "UNK" category
Flows	28 of 1,615 clients absent	Investigated (below)	Filled with 0; added <code>no_flow_data</code> flag
Targets	646 of 1,615 clients (40%)	By design as this is the targeting set	N/A

Flow imputation (investigation) The 28 clients without flow records were compared against the rest on demographics, product ownership, and (weakly, given only 18 labeled) sale rates:

- Demographics were similar (median age 46 compared to 41, tenure 108 compared to 96 months, current-account balance €407 compared to €465), not indicative of a data extraction gap.
- Product ownership was starkly different: **0% held a savings account, overdraft, credit card, or consumer loan**, versus 26–27% (SA, OVD) and 8–11% (CC, CL) in the rest of the base. Under a null hypothesis of no difference, there is not enough statistical evidence to discard the null hypothesis.
- Sale rates among the 18 labeled clients in this group were lower for MF (0/18) and CL (1/18) than the base rates (~20%, ~30%), though the sample is too small to draw a firm conclusion alone.

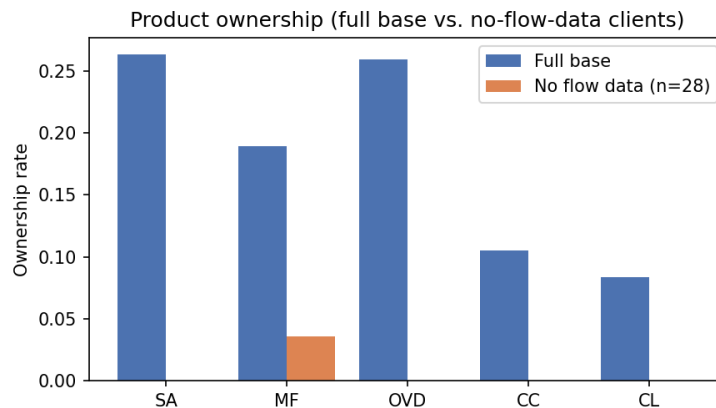


Figure 3: Ownership Distribution, missing flow client data vs remaining base

Figure 3. Regarding the ownership only a small share of the missing flow data reported ownership for Mutual Fund, the other products where not owned by this segment.

Conclusion: this is a group of mono-product, low-engagement clients rather than a data extraction failure, so flow variables were filled with 0 rather than the median. A median-fill run was tested as a sensitivity check and produced materially the same cross-validated model performance.

3. Feature engineering

The following features were implemented in `../src/features.py` as a `FlowImputer` (fit only on training folds) followed by a stateless `FeatureBuilder`:

- **Demographics:** age, tenure in years, sex indicator.
- **Product holdings:** raw counts, ownership flags, and `n_products` (total products held).
- **Balances and flows:** signed-log transform ($\text{sign}(x) \cdot \log_{10}(|x|)$) applied to all balance and flow columns, which are heavily skewed and can be negative (liabilities, net flow)

- **Derived ratios:** cashless/cash/payment-order share of debit volume, current-account share of credit volume, and average transaction size

Note: All target-related columns and `Client` are explicitly excluded from the feature matrix

4. Leakage check

Before using product-holding features, as the product snapshot predates the sales and sales occur among both existing owners and non-owners of each product.

(261 of 890 CL sales are among non-owners, and product ownership does not correlate perfectly with the sale label)

Which confirms owners should remain eligible for offers rather than being excluded a priori

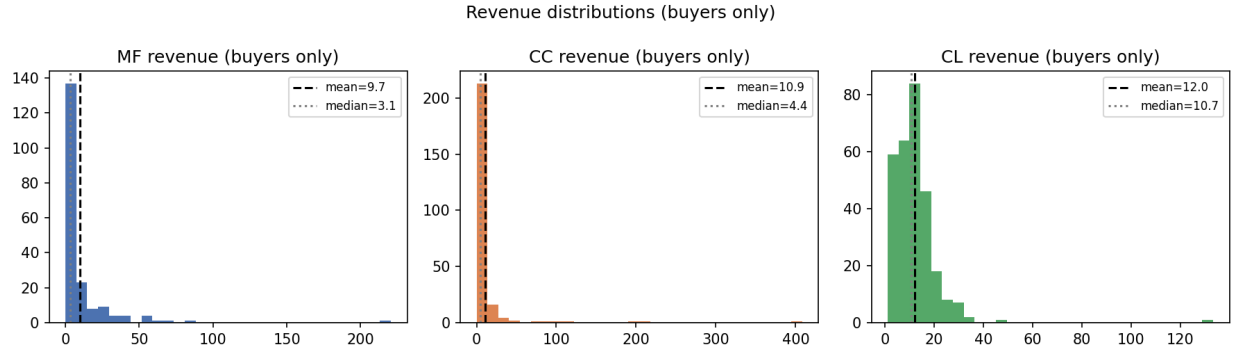


Figure 4: Revenue Distributions per Product (buyers only)

Figure 4. As seen in the buyer’s revenue distribution, in all of the studied products, there is a prominent right tail which pushes the mean, and thus implies a higher predictability effect on the median

5. Propensity models

Setup: for each product, a logistic regression (L2-regularized, standardized features) and a `HistGradientBoostingClassifier` (max depth 3, learning rate 0.05, min leaf 20, L2 regularization) were trained and evaluated with 5-fold stratified cross-validation, averaged over 5 random seeds to reduce split-dependent noise

The final propensity score is the unweighted average of the two models’ predicted probabilities (“blend”)

The blend outperformed both individual models on every product, on both AUC and top-15% lift, even though individual differences were often within one standard error:

Product	Model	AUC (mean $\pm$ std)	Lift @ 15% (mean $\pm$ std)
MF	logit	0.592 $\pm$ 0.018	1.634 $\pm$ 0.144
MF	hgb	0.580 $\pm$ 0.009	1.738 $\pm$ 0.038
MF	blend	0.599 $\pm$ 0.012	1.925 $\pm$ 0.111
CC	logit	0.583 $\pm$ 0.009	1.624 $\pm$ 0.116
CC	hgb	0.595 $\pm$ 0.009	1.817 $\pm$ 0.077
CC	blend	0.605 $\pm$ 0.007	1.916 $\pm$ 0.080
CL	logit	0.646 $\pm$ 0.005	1.880 $\pm$ 0.099
CL	hgb	0.633 $\pm$ 0.005	1.968 $\pm$ 0.066
CL	blend	0.651 $\pm$ 0.003	2.028 $\pm$ 0.081

Base rates of MF 19.9%, CC 25.0%, CL 29.9%, with 969 labeled clients and 184–290 positives per product,

AUC differences below ~ 0.02 are not statistically significant, the blend was chosen because it won consistently across all three products and both metrics, not because any single comparison was individually significant

6. Calibration

Because expected revenue is $P(\text{buy}) \times E[\text{Revenue} \mid \text{buy}]$, calibration of the propensity scores matters more than ranking quality alone. Out-of-fold blend predictions were checked against observed rates in 5 quantile bins per product, plus an overall calibration slope

- **Mean predicted probability matched the base rate exactly** for all three products (for CL 0.297 predicted vs. 0.299 observed)
- Calibration slopes (0.80–0.86) were within ~ 1 standard error of 1 and not statistically different from perfect calibration given the sample size
- **The top quintile was well calibrated in all three products** (for CL 0.527 predicted vs. 0.552 observed).
- Under-prediction appeared in the bottom quintile of all three products, consistently enough to be a real pattern, but since this range is never contacted, no recalibration was applied

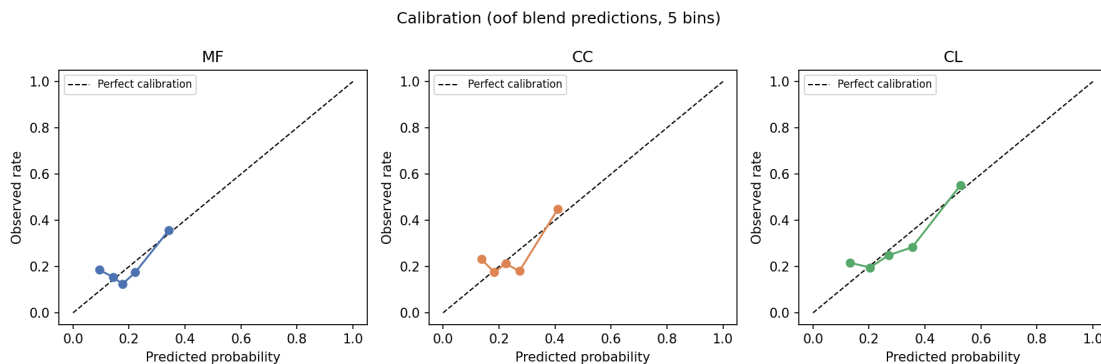


Figure 5: Calibration Curves

Figure 5. Calibration on top quintile for all 3 product, matches the expected calibration line

7. Revenue modeling

Not every client contacted will buy, so the expected revenue of an offer is split into two parts:

Expected revenue = $P(\text{buy}) \times \text{average revenue per buyer}$

The first part, the probability that a client buys, comes from the propensity models in section 5. The second part, how much a buyer typically brings in, comes from the historical buyers in the labeled data.

Buyers do not all bring in similar amounts as for mutual funds and credit cards, a small group of big buyers accounts for a large share of the total, this is why the mean is much higher than the median for these two products

Product	Mean (buyers)	Median (buyers)	Share of revenue from top 5% of buyers
MF	9.7	3.1	38%
CC	10.9	4.4	54%
CL	12.0	10.7	15%

Consumer loans are the most stable product as the mean and median are close, and no small group dominates. Credit cards are the least stable as over half of their revenue comes from about 1 in 20 buyers.

It was checked whether client characteristics (age, balances, activity, etc.) help predict revenue among buyers, using rank correlation (Spearman) and the strongest relationship found was 0.18 (consumer loan revenue vs. current-account balance), which is not strong enough implying that a value this size can easily appear by chance.

Because no feature predicts revenue reliably, fitting a revenue model would mostly fit noise. Therefore it was used one flat average revenue per buyer for each product, so that differences between clients come only from how likely they are to buy.

Since a few large buyers pull the averages up, the analysis was repeated with buyer revenues capped at the 99th percentile, which limits the influence of extreme values (average revenue per buyer became MF 9.7 $\rightarrow$ 8.9, CC 10.9 $\rightarrow$ 9.5, CL 12.0 $\rightarrow$ 11.6).

The offer mix changed somewhat (credit-card offers dropped from 27 to 16 in the back-test), but the realized back-test revenue was practically unchanged (median difference +18, 95% CI $[-28, +100]$, which includes zero). The strategy therefore does not depend on a handful of unusually large buyers.

8. Targeting strategy and validation

For each client, the expected revenue for each of the three offers ($P(\text{buy}) \times \text{mean revenue}$) is computed, and then the client is assigned to their single best offer, then the **top 15%** clients are selected by that expected value and one offer per client is then assigned

Back-test (labeled set, n=145, 15% of 969, using out-of-fold predictions):

Strategy	Realized revenue	Per contact
Full mix (CL/CC/MF optimizer)	959.8	6.62
Random clients, random offer	391.7	2.70
Random clients, all offered CL	523.4	3.61
CC + CL only (no MF eligible)	1,005.6	6.94
CL-only (rank by CL propensity)	1,154.6	7.96

Model-expected revenue for the selected 145 was 1,005.2 vs. 959.8 realized (-4.5%), consistent with a modest winner’s curse effect from selecting on noisy estimates.

CL and MF realized close to their expected values (buyer counts matched or exceeded predictions) but CC realized well below expectation (90.2 vs. 177.0), driven by the heavy right tail of CC revenue not materializing in this particular sample of 18 buyers, the standard error on that sample’s mean revenue is large enough that this gap is not statistically significant.

Strategy comparison (paired bootstrap over 1,000 resamples, same predictions):

Comparison	Median difference	95% CI
CL-only – full mix	+165	$[-14, +503]$
CL-only – CC+CL	+144	$[-33, +483]$
Winsorized full – full	+18	$[-28, +100]$

The CI for CL-only vs. the full mix narrowly includes zero, which then implies the decision to retain the full three-product optimizer as the primary strategy, as it treats all three products consistently, and the CL-only advantage rests on a small, plausibly noisy sample of CC outcomes, but the CL-only strategy can be documented/reported as a lower-variance alternative

Strategy minus “random clients, all offered CL” has a 95% CI of **[281, 654]**, which **excludes zero**, this is the strongest evidence that the propensity models add real value beyond simply offering the “highest average revenue” product to everyone.s

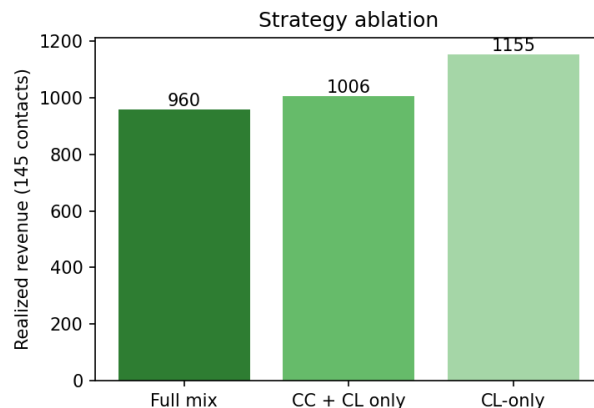


Figure 6: Strategy Ablation (visual representation of the table above)

Figure 6. The full three-product optimizer (each client gets their single best offer, and the top 15% by expected revenue are contacted), kept as the primary strategy because the CL-only alternative’s higher back-test revenue is not statistically significant

9. Final targeting list

The blended models were refit on all 969 labeled clients and applied to the 646 targeting clients. Mean predicted probabilities on the targeting set matched the labeled base rates almost exactly (CL: 0.302 vs. 0.299), confirming the two populations are comparable.

- **97 clients selected** (15% of 646, rounded from 96.9): 73 CL offers, 24 CC offers, 0 MF offers.
- **Expected revenue:** model-expected total 682.5; excluding the 10 CL contacts who already hold a consumer loan reduces this to 668.7, a minor effect, consistent with the leakage check in #4.
- **Stability checks:** list overlap with alternative strategies was 90/97 (winsorized revenue), 87/97 (excluding current product owners), and 75/97 (CL-only), the core of the list is stable across reasonable methodological choices.
- The realized-revenue back-test (available in #8) is the basis for the headline projection of 640 (range 530–790) rather than the model-expected 682.5, since it reflects actual outcomes rather than model estimates.

Outputs: - `workspace/outputs/clients_to_contact.csv`: Client, Offer, P(buy), Expected Revenue, for the 97 selected clients. - `workspace/outputs/all_targeting_clients_scored.csv`: Full scores for all 646 targeting clients (shows which clients have higher propensity, independently of the contact-budget cutoff).

10. Driver analysis

Group-wise permutation importance (5-fold CV, AUC drop, feature groups: demographics, products owned, balances, flow volumes, flow transactions) combined with a profile comparison of the top 15% of scored clients vs. the rest:

Product	Leading driver(s)	Profile of top 15%
CL	Demographics (age + tenure)*	Median age 27 compared to 45; tenure 160 compared to. 81 months; lower CA balance; higher debit activity; more existing CL holders (15% compared to 7%)

Product	Leading driver(s)	Profile of top 15%
CC	Balances, mainly savings holdings	Savings account 60% compared to 22%; CA balance 1,135 compared to 461; older (46 compared to 40)
MF	Flow transactions/volumes, products owned	Existing MF holders 44% compared to 15%; higher debit transaction count and credit turnover; lower CA balance

*by far the largest single effect (0.128 AUC drop)

An age-band compared to an “age at account opening” decomposition for CL found both bands show a similarly elevated sale rate (25 years old: 47%; joined 18 years old: 49%, compared to a 30% base rate), so the effect is described as “young, long-tenured clients” rather than attributed to a specific joining-age segment, since the two variables are collinear and cannot be separated with this data.

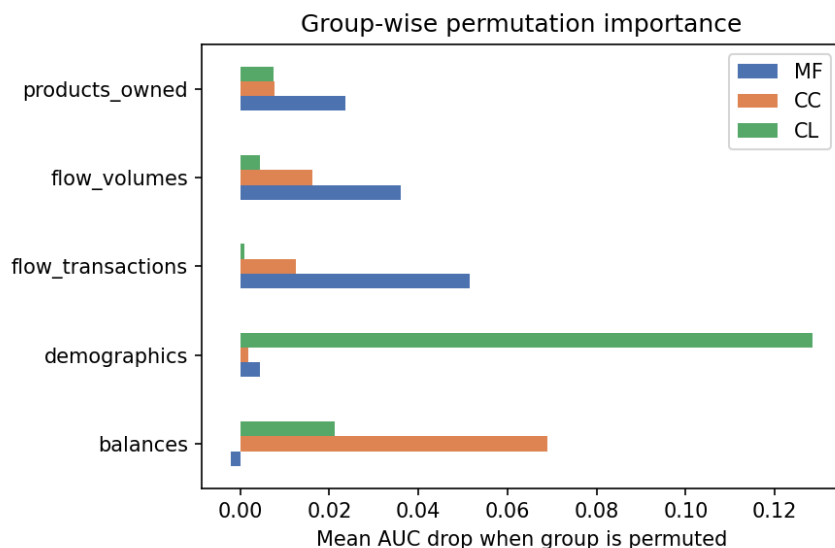


Figure 7: Feature Importance of the final models

Figure 7. As stated above, demographics were the main driver for Consumer Loans, balances were the main driver for Credit Cards and Mutual Funds were distributed along flow and ownership features

11. Assumptions and limitations

- Historical `Sale_*` labels are treated as a proxy for response to a future offer, there is no treatment/control indicator in the data, forcing a propensity model nature, not a causal uplift model
- Revenue means are estimated on the same 969 labeled clients used for back-testing, so both the revenue estimates and the back-test carry a small optimistic bias
- CC revenue is highly concentrated (54% from the top 5% of buyers), making CC-specific revenue and back-test estimates the least stable component of the analysis
- Confidence intervals reported reflect resampling uncertainty in client outcomes, as they do not capture uncertainty from refitting the models themselves
- The evidenced class imbalance is mild (19.9–29.9% positive rates), so no class-balancing technique was applied

12. Reproducibility

- **Python:** 3.x (pinned versions in `requirements.txt`)
- **Key packages:** pandas, numpy, scikit-learn (`LogisticRegression`, `HistGradientBoostingClassifier`), scipy
- **Random seeds:** CV results averaged over 5 seeds (0–4) `StratifiedKFold(shuffle=True, random_state=seed)`
- **CV scheme:** 5-fold stratified cross-validation on the 969 labeled clients, repeated per seed, no additional hyperparameter tuning beyond what's specified in `../src/models.py`
- **Final models:** refit on 100% of the labeled set using the same fixed hyperparameters used during CV
- **To reproduce:** `pip install -r requirements/used.txt`, then run the notebook below
- **Notebook:** `/workspaces/kbc/workspace/notebooks/dev.ipynb`

Appendix: repository structure

```
data/raw/    data/processed/master.parquet
src/features.py      # FlowImputer, FeatureBuilder
src/models.py        # make_model, cv_eval, lift_at
notebooks/01_eda.ipynb
outputs/clients_to_contact.csv
outputs/all_targeting_clients_scored.csv
reports/executive_summary.pdf
reports/technical_report.md
```

Prepared by Alejandro Gonzalez, September 2026, for the KBC Data Scientist case study.